



Java SPRING BOOT

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INTRODUCTION TO SPRING BOOT

• WHAT IS SPRING BOOT?

- A JAVA-BASED FRAMEWORK USED TO CREATE STAND-ALONE, PRODUCTION-READY APPLICATIONS.
- REDUCES THE NEED FOR BOILERPLATE CODE AND CONFIGURATIONS.

• WHY SPRING BOOT?

- EASY TO SET UP AND GET STARTED.
- GREAT FOR BUILDING REST APIS AND MICROSERVICES.

KEY FEATURES OF SPRING BOOT

SPRING BOOT PROVIDES AUTO-CONFIGURATION AND STARTER DEPENDENCIES TO REDUCE SETUP TIME. IT AUTO-DETECTS LIBRARIES AND CONFIGURES THEM BASED ON WHAT'S PRESENT IN THE CLASSPATH.

THE APPLICATION STARTS WITH `@SPRINGBOOTAPPLICATION` AND RUNS ON EMBEDDED SERVERS LIKE TOMCAT. THIS ALLOWS DEVELOPERS TO BUILD AND RUN SPRING APPS AS STANDALONE JARS.

IT SUPPORTS A LAYERED ARCHITECTURE WITH CLEAR SEPARATION INTO CONTROLLER, SERVICE, AND REPOSITORY. ANNOTATIONS LIKE `@RESTCONTROLLER` AND `@SERVICE` DEFINE ROLES CLEARLY AND KEEP THE CODE CLEAN.

JSP OVERVIEW

JAVA SERVER PAGES (JSP) IS A TECHNOLOGY THAT HELPS IN CREATING DYNAMIC WEB PAGES. IT ALLOWS JAVA CODE TO BE EMBEDDED DIRECTLY INTO HTML PAGES.

JSP Lifecycle: From Request to Destruction

Translation Phase

JSP file translated into Java servlet



Request Handling Phase

Servlet processes incoming requests



Destroy Phase

Servlet destroyed and resources cleaned up



Compilation Phase

Servlet compiled into bytecode



Initialization Phase

Servlet initialized and placed in memory



JSP

● JSP EXPRESSIONS

EMBED JAVA CODE IN HTML. THE RESULT IS INSERTED INTO THE PAGE DYNAMICALLY DURING REQUEST PROCESSING.

● JSP DECLARATIONS

DECLARE VARIABLES OR METHODS INSIDE THE JSP, MAKING THEM AVAILABLE THROUGHOUT THE PAGE.

JSP



JSP EXPRESSIONS ARE USED TO EMBED JAVA CODE DIRECTLY INTO HTML, ALLOWING DYNAMIC CONTENT GENERATION. THEY EVALUATE JAVA EXPRESSIONS AND INSERT THE RESULT INTO THE RESPONSE SENT TO THE CLIENT.



JSP ACTIONS ARE SPECIAL TAGS USED TO CONTROL THE BEHAVIOR OF THE JSP CONTAINER OR INTERACT WITH OTHER RESOURCES. THEY SIMPLIFY TASKS LIKE INCLUDING CONTENT, FORWARDING REQUESTS, AND MANAGING JAVABEANS WITHIN A JSP PAGE.



JSP

JSP DIRECTIVES

JSP DIRECTIVES PROVIDE CONFIGURATION FOR THE PAGE, SUCH AS LANGUAGE, CONTENT TYPE, AND ERROR HANDLING. THEY AFFECT THE PROCESSING BEHAVIOR OF THE JSP CONTAINER.

JSP ACTIONS

JSP ACTIONS ARE TAGS THAT CONTROL THE JSP ENGINE'S BEHAVIOR, SUCH AS INCLUDING FILES, FORWARDING REQUESTS, AND WORKING WITH JAVABEANS. THEY SIMPLIFY RESOURCE MANAGEMENT AND CODE.

JSP IMPLICIT OBJECTS

IMPLICIT OBJECTS ARE PREDEFINED OBJECTS PROVIDED BY THE JSP CONTAINER, ALLOWING EASY ACCESS TO REQUEST PARAMETERS, SESSION DATA, AND OUTPUT STREAMS WITHOUT EXPLICIT DECLARATIONS.

EXPRESSION LANGUAGE

EXPRESSION LANGUAGE (EL)

**SIMPLIFIES ACCESSING
DATA IN JSP PAGES
WITHOUT USING JAVA
CODE. IT RETRIEVES
VALUES FROM JAVABEANS,
COLLECTIONS, MAPS, OR
OTHER OBJECTS IN A
STRAIGHTFORWARD
WAY REDUCES THE NEED
FOR BOILERPLATE CODE
AND CONFIGURATIONS.**

BENEFITS OF EL

**PROVIDES CLEANER,
MORE READABLE
CODE AND ENSURES
BETTER SEPARATION
OF BUSINESS LOGIC
FROM THE
PRESENTATION
LAYER.**

RMI PROCESS

Define Remote Interface

The interface is created to specify remote methods.



Client Calls Method

The client invokes a method on the remote object.



Server Responds

The server processes the request and sends back the result.



Register Remote Object

The remote object is registered with the RMI registry.



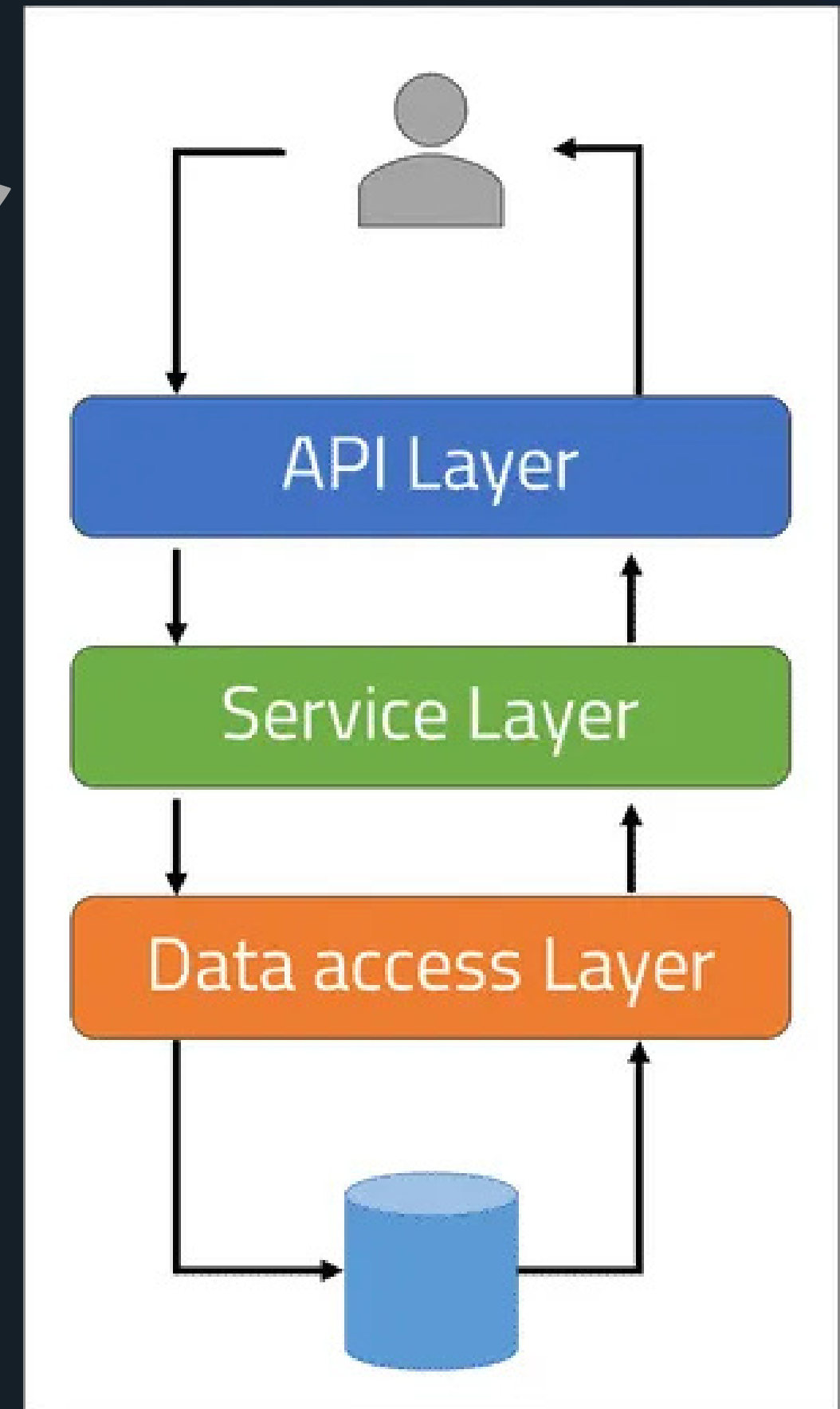
Network Connection

The network facilitates communication between client and server.



SPRING MVC 3-LAYER ARCHITECTURE

SPRING MVC USES A 3-LAYER ARCHITECTURE: CONTROLLER, SERVICE, AND REPOSITORY. THE CONTROLLER HANDLES USER INPUT, THE SERVICE PROCESSES BUSINESS LOGIC, AND THE REPOSITORY INTERACTS WITH THE DATABASE. THIS STRUCTURE ENSURES CLEAN SEPARATION OF CONCERNS AND EASY MAINTENANCE.



REST API IN JAVA

WHAT IS REST?

**REST IS A SIMPLE,
STATELESS
ARCHITECTURE USING
HTTP METHODS LIKE
GET, POST, PUT, AND
DELETE TO MANAGE
RESOURCES OVER THE
WEB**

WHY USE REST IN JAVA?

**IT'S LIGHTWEIGHT,
EASY TO INTEGRATE
WITH FRONTENDS,
AND IDEAL FOR
BUILDING SCALABLE
APIS FOR WEB AND
MOBILE
APPLICATIONS.**

SPRING BOOT & REST

**SPRING BOOT SIMPLIFIES
REST API CREATION
USING ANNOTATIONS. IT
HANDLES HTTP
REQUESTS AND
RESPONSES EFFICIENTLY,
RETURNING DATA IN
JSON FORMAT WITH
MINIMAL SETUP.**

EJB

Security

Measures that protect beans and data from unauthorized access



Session Beans

Beans that manage client sessions and business logic



Transactions

Mechanisms that ensure data integrity and consistency



Entity Beans

Beans that represent persistent data and database interactions



Lifecycle Control

Processes that manage the creation and destruction of beans



Message-Driven Beans

Beans that handle asynchronous messaging and event processing



SPRING FRAMEWORK OVERVIEW

SPRING FRAMEWORK INTRODUCTION:

**A COMPREHENSIVE
FRAMEWORK FOR
JAVA APPLICATIONS,
SUPPORTING DI, AOP,
AND SPRING MVC FOR
WEB APPS.**

DEPENDENCY INJECTION (DI):

**ALLOWS AUTOMATIC
INJECTION OF
DEPENDENCIES,
PROMOTING LOOSE
COUPLING.**

ASPECT-ORIENTED PROGRAMMING (AOP)

**SEPARATES CROSS-
CUTTING CONCERNS
LIKE LOGGING AND
TRANSACTION
MANAGEMENT FROM
BUSINESS LOGIC.**

HIBERNATE FRAMEWORK OVERVIEW

HIBERNATE INTRODUCTION

**HIBERNATE
SIMPLIFIES DATABASE
INTERACTIONS BY
MAPPING JAVA
OBJECTS TO
DATABASE TABLES
USING ORM.**

HIBERNATE ARCHITECTURE

**WORKS WITH
SESSION FACTORIES
AND SESSIONS FOR
DATABASE
OPERATIONS,
REDUCING THE NEED
FOR SQL.**

(HQL)

**A QUERY
LANGUAGE FOR
INTERACTING WITH
DATABASE
OBJECTS INSTEAD
OF TABLES.**

INVERSION OF CONTROL (IOC)

INVERSION OF CONTROL (IOC) IS A DESIGN PRINCIPLE WHERE THE CONTROL OF OBJECT CREATION AND DEPENDENCY MANAGEMENT IS GIVEN TO THE SPRING CONTAINER, INSTEAD OF BEING HANDLED MANUALLY BY THE PROGRAMMER.

BENEFITS

- **LOOSE COUPLING**
- **BETTER TESTABILITY**
- **CLEANER AND MAINTAINABLE CODE**



Dependency Injection



Automatic Object Provision



Annotation Usage

CERTIFICATE OF EXCELLENCE

THIS CERTIFICATE IS AWARDED TO

SCALER
Topics

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In recognition of the completion of the tutorial: **Spring Boot Course: Certified Course for Essential Skills**

Following are the the learning items, which are covered in this tutorial

▶ 18 Video Tutorials 🔒 4 Modules

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THANK
YOU

